

M2R: Options Pricing

Harry Zheng
email: h.zheng@imperial.ac.uk
office: Room 701, Weeks Building

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Objectives. This M2R is to use and compare different methods (closed-form formula, binomial tree, finite difference, and simulation) to price European call options. The first part is a simple Black-Scholes model (Week 1) and the second part is an extension to multi-asset model or stochastic volatility model (Week 2 and 3). If time permits, one can also explore the deep neural network method.

References. The following books are loanable at Imperial College Library and available for online access.

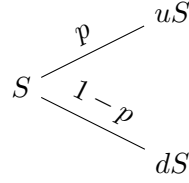
1. J.C. Hull (2021), *Options, Futures, & Other Derivatives*, 11th edition, Pearson.
2. Y.K. Kwok (2008), *Mathematical Models of Financial Derivatives*, 2nd edition, Springer.
3. P. Wilmott, S. Howison, J. Dewynne (1995), *The Mathematics of Financial Derivatives – A Student Introduction*, Cambridge University Press.

1 Binomial Model

Concepts in option pricing theory. A *call* (or *put*) option is a contract that gives the holder the *right* to buy (or sell) a prescribed asset (*underlying asset*) by a certain date T (*expiration date*) for a predetermined price K (*exercise price*). A *European* option can only be exercised on the expiration date. The other party to the holder of the option contract is called the *writer*. The holder and the writer are said to be in *long* and *short* positions of the contract, respectively. The *terminal payoffs* of a European call (or put) option is $(S(T) - K)^+ := \max(S(T) - K, 0)$ (or $(K - S(T))^+$), where $S(T)$ is the underlying asset price at time T .

One period model. Assume risky asset price is S_0 at beginning of the period (time 0). At the end of the period (time 1), asset price may go up to $S_1(H)$ (head of tossing a coin) or go down to $S_1(T)$ (tail of tossing a coin). We normally write $S_1(H) = uS_0$ and $S_1(T) = dS_0$, where $0 < d < u$. Outcome of tossing a coin is a binomial random variable taking value H with probability p and T with probability $1 - p$, $0 < p < 1$, p is a real (empirical, physical) probability.

There is riskless asset in the market, called *cash bond* or *savings account* or *money market*, normally offered by banks. If one deposits (or borrows) 1 at beginning of the period, then one receives (or repays) $1 + r$ at the end of the period, $r \geq 0$ is called *riskless interest rate*.



An essential feature of an efficient market is that if a trading strategy can turn nothing into something, then it must also run the risk of loss. Otherwise, there would be an *arbitrage*.

Definition 1 (Arbitrage). *An arbitrage is a portfolio (a trading strategy, or an investing strategy) L with zero initial capital and with final value V_T^L which satisfies $V_T^L \geq 0$ a.s. (almost surely) (i.e. $\mathbb{P}(V_T^L \geq 0) = 1$) and $V_T^L > 0$ with strictly positive probability (i.e. $\mathbb{P}(V_T^L > 0) > 0$).*

Whenever we model a market, we should choose a model that is arbitrage-free (i.e. admits no arbitrage). Indeed, an arbitrage is an unrealistic strategy, too good to be allowed to exist: while in the real world arbitrage fleetingly occurs, some traders (called arbitrageurs) quickly notice the arbitrage and take advantage of it and drive the prices back to equilibrium. Thus, we should exclude as nonsensical all models in which there exists an arbitrage.

Standing assumption. *Market is arbitrage free.* This is fundamental requirement, without it, we will not be able to do options pricing.

No arbitrage condition. We have the following relation:

$$0 < d < 1 + r < u.$$

If $1 + r \geq u$, then one can *short sell* (meaning that the investor does not own the stock, but borrows the share from a bank or a broker and sells it in the market, then at a later time buys it from the market and returns it to the lender.) one unit of risky asset with price S at time 0, deposit S in the bank, giving a wealth value at time 0 as

$$X_0 = 0.$$

The risky asset price at time 1 is S_1 which takes value uS or dS . The wealth at time 1 is

$$X_1 = (1 + r)S - S_1 \geq (1 + r)S - uS \geq 0,$$

no matter “up” or “down” state of risky asset S . Remember S_1 and X_1 are random variables depending on outcome of coin tossing. Moreover, in the “down” state (with probability $1 - p > 0$) the wealth value is

$$X_1 = (1 + r)S - S_1 = (1 + r)S - dS \geq uS - dS > 0.$$

Hence we can make a riskless profit with positive probability, which contradicts no arbitrage assumption. Therefore, $u > 1 + r$. (In the argument above, we have assumed that we can short sell risky asset, in the real market this may not be possible.) A similar argument yields that $d < 1 + r$.

Replication pricing principle. The derivative’s price equals the value of replicating portfolio.

The idea is simple: if one can find a portfolio that replicates the payoff of option at time 1, irrespective of “up” or “down” state, then the option value at time 0 must equal to the replicating portfolio value at time 0, otherwise, there is arbitrage.

Replicating portfolio. We form a portfolio consisting of Δ_0 units of underlying asset ($\Delta_0 > 0$ buying, $\Delta_0 < 0$ short selling) and cash amount B in riskless cash bond ($B > 0$ lending, $B < 0$ borrowing). Asset price at time 0 is $S_0 = S$. The payoff at time 1 is a function of S_1 , e.g., $V_1 = (S_1 - K)^+$ for call option, $V_1 = (K - S_1)^+$ for put option, $V_1 = S_1 - K$ for forward contract.

Replicating portfolio (or wealth) value at time 0 is $X_0 = \Delta_0 S + B$, and at time 1 is

$$X_1 = \Delta_0 S_1 + RB = \begin{cases} \Delta_0 uS + RB =: X_1(H) & \text{w. prob. } p \\ \Delta_0 dS + RB =: X_1(T) & \text{w. prob. } 1 - p, \end{cases}$$

where $R = 1 + r$. Option value at time 1 is given by V_1 which is a random variable, taking value $V_1(H)$ with probability p and $V_1(T)$ with probability $1 - p$. To replicate the option, we must have $X_1(H) = V_1(H)$ and $X_1(T) = V_1(T)$, irrespective probability p . Hence

$$\begin{cases} \Delta_0 uS + RB = V_1(H) \\ \Delta_0 dS + RB = V_1(T) \end{cases} \quad (1)$$

Solving the equation, we have

$$\Delta_0 = \frac{V_1(H) - V_1(T)}{uS - dS}, \quad B = \frac{uV_1(T) - dV_1(H)}{uR - dR}$$

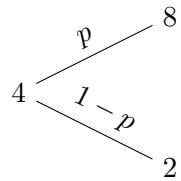
The arbitrage free price for the option at time 0 is given by $V_0 = X_0 = \Delta_0 S + B$, that is,

$$V_0 = \frac{1}{R} [qV_1(H) + (1 - q)V_1(T)], \quad (2)$$

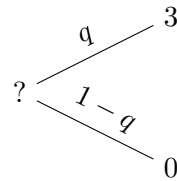
where $q = \frac{R-d}{u-d}$. Since $d < R < u$, we have $0 < q < 1$. We can introduce a probability measure Q such that $Q(H) = q$ and $Q(T) = 1 - q$, such Q is called the *risk neutral probability measure* or *equivalent martingale measure (EMM)*. (2) is called the *risk neutral pricing formula*.

Example 1. Assume $S_0 = 4$ at time 0, $u = 2, d = 1/2$ and $r = 1/4$. Consider a European call option with exercise price $K = 5$ and exercise time 1. Find call price and trading strategy at time 0, also show replicating portfolio produces the same value as call option at time 1.

Solution.



Asset Tree



Derivative tree

(replication equation) Setting $X_1 = RB + \Delta_0 S_1$ equal to V_1 gives the following system

$$\frac{5}{4}B + 8\Delta_0 = 3, \quad \frac{5}{4}B + 2\Delta_0 = 0$$

whose solution is $B = -\frac{4}{5}, \Delta_0 = \frac{1}{2}$, which corresponds to the initial cash $B + \Delta_0 S_0 = -\frac{4}{5} + \frac{1}{2} \cdot 4 = \frac{6}{5}$. At time 1, the value of the replicating portfolio is $RB + \Delta_0 S_1 = -1 + \frac{1}{2} S_1$, which is equal to 3 if

the asset price goes up and 0 if the asset price goes down. So the replicating portfolio produces the same value as the call option at time 1. $\square$

Two period model. Asset price at time 0 is $S_0 = S$, at time 1 (end of first period), asset price S_1 is a random variable that equals uS if H occurs and dS if T occurs. In case of $S_1 = uS$, then at time 2 (end of second period), asset price S_2 is a random variable that equals $u(uS)$ if H occurs and $d(uS)$ if T occurs. On the other hand, In case of $S_1 = dS$, then at time 2 (end of second period), asset price S_2 is a random variable that equals $u(dS)$ if H occurs and $d(dS)$ if T occurs. Therefore, S_2 is a random variable that can take 3 possible values u^2S, udS, d^2S .

The value of derivative (option) V_0 at time 2 is a random variable, taking one of three values, depending on S_2 . To find its value at time 0 and use one-period formula, we would need to know its values at time 1, that is, V_1 , which is random variable, depending on S_1 , but can be computed with one period formula and values of $V_2(S_2)$ at time 2. Specifically,

$$\begin{aligned} V_1(H) &= \frac{1}{R} \left(qV_2(HH) + (1-q)V_2(HT) \right) \\ V_1(T) &= \frac{1}{R} \left(qV_2(TH) + (1-q)V_2(TT) \right). \end{aligned}$$

The replicating strategy at time 1 is given by

$$\Delta_1(\omega_1) = \frac{V_2(\omega_1 H) - V_2(\omega_1 T)}{S_2(\omega_1 H) - S_2(\omega_1 T)},$$

where ω_1 equals H or T , depending on coin tossing at end of period 1, which is also a random variable depending on ω_1 . We can then compute the value V_0 at time 0 by (2).

In summary, we can use backward method and one period formula to compute two-period value of derivatives at time 0 and the intermediate arbitrage-free values and replication trading strategies at time 1 are determined based on the information (filtration) of stock price at time 1.

Multi-period model. We can continue this process and build up asset price tree for multi-periods until the maturity of the option, say time N , and then use backward method and one period formula to find the value of the derivative at time 0. Note the asset price tree built in this way is recombining, that is, up-down moves and down-up moves would lead to the same point, so the total number of nodes at time n is $n + 1$. If tree is not recombining, then the total number of nodes at time n is 2^n , too big to be stored and computed if n is large, which is called *curse of dimensionality*.

Implementation. Assume the expiration (exercise) time is n and derivative payoffs only depend on risky asset values at time n , i.e., they are path independent. At time i there are $i + 1$ nodes. Denote these nodes as (i, j) with $i = 0, 1, \dots, n$ and $j = 0, 1, \dots, i$, where the first component i refers to time i and the second component j refers to the asset price $S_j^i := Su^j d^{i-j}$. Compute the option prices V_j^n , $j = 0, 1, \dots, n$, at expiration time n over $n + 1$ nodes. Then go backwards one step and compute option prices at time $n - 1$ using the one period binomial option pricing formula over n nodes. Continue until reaching time 0. The recursive formula is

$$V_j^i = \frac{1}{R} \left(qV_{j+1}^{i+1} + (1-q)V_j^{i+1} \right), \quad j = 0, \dots, i \text{ and } i = n - 1, \dots, 0,$$

where $q = \frac{R-d}{u-d}$. The value V_0^0 is the option price at time 0. The hedging (or replicating portfolio) strategy at node (i, j) is

$$\Delta_j^i = \frac{V_{j+1}^{i+1} - V_j^{i+1}}{S_{j+1}^{i+1} - S_j^{i+1}}, \quad j = 0, \dots, i \text{ and } i = n - 1, \dots, 0.$$

2 Limit of Discrete Time Models

Consider time interval $[0, T]$, where T is maturity time for option, and 0 initial time. Assume $t \in [0, T]$ is current time, then $T - t$ is time to remain to maturity. We may divide interval $[t, T]$ by N equally spaced sub-intervals with step size $\Delta t = (T - t)/N$ and grid points $t_i = t + i\Delta t$, $i = 0, 1, \dots, N$. So $t_0 = t$ and $t_N = T$. This is the multi-period model with $S_0 = S(t_0) = S(t)$, $S_1 = S(t_1) = S(t + \Delta t)$, etc. The interest rate over period $[t, t + \Delta t]$ is $1 + r_{\Delta t}$, we often use continuously compounding interest rate, which would be $e^{r_{\Delta t}}$, where r annual interest rate, so $r_{\Delta t} = e^{r_{\Delta t}} - 1$. All results we have derived can be applied here, just remember that $S_n = S(t_n) = S(t + n\Delta t)$, etc.

In a risk neutral continuous time model (to be discussed in detail later), over a period $[t, t + \Delta t]$ the asset price $S(t + \Delta t)$ can be expressed in terms of $S(t) = S$ as

$$S(t + \Delta t) = S e^{(r - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}Z} \quad (3)$$

where r is the riskless interest rate, σ the volatility, and $Z \sim N(0, 1)$ a standard normal random variable. $S(t + \Delta t)$ is a lognormal random variable, that is, $\ln S(t + \Delta t)$ is a normal variable. We have $S(t + \Delta t)$ has the first moment $S e^{r\Delta t}$ and the second moment $S^2 e^{(2r + \sigma^2)\Delta t}$.

By equating the first and second moments of the asset price $S(t + \Delta t)$ in both continuous and discrete time models, and cancelling S on both sides, we obtain the relation

$$q u + (1 - q) d = e^{r\Delta t}, \quad (4)$$

$$q u^2 + (1 - q) d^2 = e^{(2r + \sigma^2)\Delta t}. \quad (5)$$

There are two equations and three unknowns u, d, q . An extra condition is needed to uniquely determine a solution. Note that (4) implies $q = \frac{R - d}{u - d}$, where $R = e^{r\Delta t}$ as before.

Cox–Ross–Rubinstein model. The extra condition is

$$u d = 1. \quad (6)$$

The approximate solutions to (4), (5), (6) are

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}, \quad q = \frac{R - d}{u - d}. \quad (7)$$

These are the parameters chosen by Cox–Ross–Rubinstein for their model. Note that with this choice (5) is satisfied up to $O((\Delta t)^2)$.

As the time interval Δt tends to zero, the one period option pricing formula, (2) with (4) and (5), tends to the Black–Scholes Equation

$$\frac{\partial V}{\partial t} + r S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - r V = 0. \quad (8)$$

Proof. Two variable Taylor expansion at (S, t) gives

$$\begin{aligned} V(u S, t + \Delta t) &= V + V_S (u S - S) + V_t \Delta t \\ &\quad + \frac{1}{2} (V_{SS} (u S - S)^2 + 2 V_{St} (u S - S) \Delta t + V_{tt} (\Delta t)^2) \\ &\quad + \text{higher order terms.} \end{aligned}$$

Similarly,

$$\begin{aligned} V(dS, t + \Delta t) &= V + V_S (dS - S) + V_t \Delta t \\ &\quad + \frac{1}{2} (V_{SS} (dS - S)^2 + 2 V_{St} (dS - S) \Delta t + V_{tt} (\Delta t)^2) \\ &\quad + \text{higher order terms.} \end{aligned}$$

Substituting into the right hand side of the one period formula (2) gives

$$\begin{aligned} V &= \frac{1}{R} \left(qV(uS, t + \Delta t) + (1 - q)V(dS, t + \Delta t) \right) \\ &= \{ V + S V_S [q(u - 1) + (1 - q)(d - 1)] + V_t \Delta t \\ &\quad + \frac{1}{2} [S^2 V_{SS} (q(u - 1)^2 + (1 - q)(d - 1)^2) \\ &\quad + 2 S V_{St} (q(u - 1) + (1 - q)(d - 1)) \Delta t \\ &\quad + V_{tt} (\Delta t)^2] + \text{higher order terms} \} e^{-r \Delta t}. \end{aligned}$$

Now it follows from (4) that

$$q(u - 1) + (1 - q)(d - 1) = qu + (1 - q)d - 1 = e^{r \Delta t} - 1 = r \Delta t + O((\Delta t)^2).$$

Similarly, (4) and (5) give that

$$\begin{aligned} q(u - 1)^2 + (1 - q)(d - 1)^2 &= qu^2 + (1 - q)d^2 - 2(qu + (1 - q)d) + 1 \\ &= e^{(2r + \sigma^2) \Delta t} - 2e^{r \Delta t} + 1 = \sigma^2 \Delta t + O((\Delta t)^2). \end{aligned}$$

So, for the RHS of (2) we get

$$\begin{aligned} V &= \{ V + S V_S (r \Delta t + O((\Delta t)^2)) + V_t \Delta t \\ &\quad + \frac{1}{2} [S^2 V_{SS} (\sigma^2 \Delta t + O((\Delta t)^2)) + 2 V_{St} (r \Delta t + O((\Delta t)^2)) \Delta t] \\ &\quad + O((\Delta t)^2) \} e^{-r \Delta t} \\ &= \{ V + (r S V_S + V_t + \frac{1}{2} \sigma^2 S^2 V_{SS}) \Delta t + O((\Delta t)^2) \} (1 - r \Delta t + O((\Delta t)^2)) \\ &= V + (-r V + r S V_S + V_t + \frac{1}{2} \sigma^2 S^2 V_{SS}) \Delta t + O((\Delta t)^2). \end{aligned}$$

Cancelling V and dividing by Δt yields

$$-r V + r S V_S + V_t + \frac{1}{2} \sigma^2 S^2 V_{SS} + O(\Delta t) = 0.$$

Ignoring the $O(\Delta t)$ term, we get the BSE. The one period binomial model approximates BSE to 1st order accuracy.

3 Continuous time models

We now introduce the Black and Scholes model in continuous time. The most important stochastic process associated with it is the Brownian motion.

Definition 2. A (standard, one dimensional) Brownian Motion (BM) is a process $W = (W_t)_{t \geq 0}$ satisfying

0. $W_0 = 0$.

1. W has independent increments, that is, if $0 \leq t_1 < t_2 \leq t_3 < t_4$, then $W_{t_2} - W_{t_1}$ and $W_{t_4} - W_{t_3}$ are independent.

2. $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $0 \leq s < t$.

3. W is continuous (i.e. $t \mapsto W_t(\omega)$ continuous $\mathbb{P}$ a.e. ω).

Sample paths of BMs. To generate a sample path of BM over interval $[0, T]$, we can do the following: Divide $[0, T]$ into n equally spaced subintervals with grid points $t_i = ih$, $i = 0, 1, \dots, n$, and $h = T/n$. Since $W_{t_i} = W_{t_{i-1}} + W_{t_i} - W_{t_{i-1}}$ and $W_{t_i} - W_{t_{i-1}} \sim \mathcal{N}(0, h)$ and all increments are independent, we can generate n independent standard normal variables Z_i , $i = 1, \dots, n$, and set

$$W_{t_i} = W_{t_{i-1}} + \sqrt{h}Z_{i-1}$$

with $W_{t_0} = 0$ for $i = 1, \dots, n$. Linking these points forms a sample path of BM.

Remark 3. A process $W = (W^i)_{i=1}^n$ with values in $\mathbb{R}^n$ is called a n -dimensional BM if its components $W^1, \dots, W^n$ are independent processes, and each component is a 1-dimensional BM.

Ito's Formula. Assume X is a stochastic process (or stochastic differential equation (SDE)) satisfying

$$dX_t = g(t, X_t)dt + \sigma(t, X_t)dW_t \quad (9)$$

with $X_0 = x$, where g, σ are well-defined functions, X_t is called *Ito process*. Assume $f \in C^{1,2}$, i.e., partial derivatives f_t, f_x, f_{xx} exist and are continuous. Then

$$\begin{aligned} df(t, X_t) &= f_t dt + f_x dX_t + \frac{1}{2} f_{xx} \sigma_t^2 dt \\ &= (f_t + f_x g_t + \frac{1}{2} f_{xx} \sigma_t^2) dt + f_x \sigma_t dW_t, \end{aligned}$$

where f_t, f_x, f_{xx} on RHS are evaluated at (t, X_t) .

Ito's formula can be derived intuitively as follows: using two variable Taylor expansion to order 2, we have

$$df(t, X_t) = f_t dt + f_x dX_t + \frac{1}{2} f_{tt} (dt)^2 + f_{tx} dt dX_t + \frac{1}{2} f_{xx} (dX_t)^2.$$

Since $dW_t \sim N(0, dt)$, we get $(dW_t)^2 = dt$, and

$$(dX_t)^2 = g_t^2 (dt)^2 + 2g_t \sigma_t dt dW_t + \sigma_t^2 (dW_t)^2 = g_t^2 (dt)^2 + 2g_t \sigma_t dt dW_t + \sigma_t^2 dt.$$

Substituting dX_t into $df(t, X_t)$ and collect terms for dt , and keep terms dt and dW_t , but discard all higher order small terms ($dt dW_t, (dt)^2$ etc), then we have Ito's formula.

Sample paths of SDE (9). To generate a sample path of X over interval $[0, T]$, we can do the following: Divide $[0, T]$ into n equally spaced subintervals with grid points $t_i = ih$, $i = 0, 1, \dots, n$, $h = T/n$, and generate n independent standard normal variables Z_i , $i = 1, \dots, n$, and set

$$X_{t_i} = X_{t_{i-1}} + g(t_{i-1}, X_{t_{i-1}})h + \sigma(t_{i-1}, X_{t_{i-1}})\sqrt{h}Z_{i-1}$$

with $X_{t_0} = x$ for $i = 1, \dots, n$. Linking these points forms a sample path of X .

4 Black Scholes Model

Model. Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, (W_t)_{t \in [0, T]})$ be Wiener space, i.e. W_t is Brownian motion, $\mathcal{F}_t$ is the natural filtration of W_t and $\mathcal{F} = \mathcal{F}_T$. A market contains two underlying securities. One is the risk-free asset (savings account) and the other risky asset. The risk-free asset is described by a deterministic function

$$dB_t = rB_t dt \quad (10)$$

with $B_0 = 1$, where r is the constant risk-free interest rate with continuous compounding. The solution is $B_t = e^{rt}$ for $t \geq 0$.

The price process of the risky asset S_t is supposed to be *geometric Brownian motion (GBM)*: S_t satisfies the SDE, for $0 \leq t \leq T$,

$$dS_t = rS_t dt + \sigma S_t dW_t, \quad (11)$$

where σ is *volatility* of S . (11) has a unique solution, for $0 \leq t \leq T$,

$$S_t = S_0 \exp\left((r - \frac{1}{2}\sigma^2)t + \sigma W_t\right). \quad (12)$$

Black-Scholes: the original proof. Black's and Scholes' original proof of the famous formula was a very direct argument showing that a replicating portfolio exists for the European call option. Here is a version of that argument.

The idea is to assume a whole lot of things and then show they are all true. The first assumption is that there is a smooth function $C(t, S)$ such that the call option has a value $C(t, S_t)$ at time $t < T$, with $\lim_{t \uparrow T} C(t, S) = [S - K]^+$. Suppose we form a portfolio in which we are long one unit of the call option and short a self-financing portfolio $(\phi, \psi, C(0, S_0))$. The value of this portfolio at time t is then

$$X_t = C(t, S_t) - \phi_t S_t - \psi_t B_t,$$

with, in particular, $X_0 = 0$. By the Ito formula and the self-financing property,

$$dX_t = \frac{\partial C}{\partial S} dS_t + \left(\frac{\partial C}{\partial t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 C}{\partial S^2} \right) dt - \phi_t dS_t - \psi_t dB_t.$$

If we choose $\phi_t = \partial C / \partial S$ and use the fact that $dB = rB dt$ we see that

$$dX_t = \left(\frac{\partial C}{\partial t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 C}{\partial S^2} - \psi_t r B_t \right) dt.$$

Let us now choose

$$\psi_t = \frac{\frac{\partial C}{\partial t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 C}{\partial S^2}}{r B_t}.$$

Then $X_t \equiv 0$, so that

$$C = \phi S + \psi B = S \frac{\partial C}{\partial S} + \frac{\frac{\partial C}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 C}{\partial S^2}}{r},$$

showing that C must satisfy the Black-Scholes PDE

$$\frac{\partial C}{\partial t} + rS \frac{\partial C}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} - rC = 0 \quad (13)$$

with terminal condition

$$C(T, S) = [S - K]^+, \quad \forall S \geq 0 \quad (14)$$

and boundary condition

$$C(t, 0) = 0, \quad \lim_{S \rightarrow \infty} C(t, S) = \infty, \quad \forall t \in [0, T]. \quad (15)$$

Solving linear PDE (13), we find call price C is given by

$$C(t, S) = S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2), \quad (16)$$

where Φ is the cumulative distribution function of a standard normal variable and

$$\begin{aligned} d_1 &= \frac{\log(S/K) + (r + \sigma^2/2)(T - t)}{\sigma\sqrt{T - t}} \\ d_2 &= d_1 - \sigma\sqrt{T - t} \end{aligned}$$

Greeks. Two most common sensitivity measures of a European call option price C are *delta*, *gamma*, defined by

$$\delta = \frac{\partial C}{\partial S} = \Phi(d_1), \quad \gamma = \frac{\partial^2 C}{\partial S^2} = \frac{\Phi'(d_1)}{S\sigma\sqrt{T - t}}.$$

Dynamic Hedging. Greeks can be used to hedge an option position against changes of underlying factors. For example, to hedge a short position of a European call against changes of the asset price, one should keep a long position of the underlying asset with δ units. This strategy requires continuous trading and is prohibitively expensive if there are any transaction costs.

Finite Difference Methods. To solve PDE (13) numerically, we may fix a range $[S_{min}, S_{max}]$, then let $S_t = S_{max}$ if S_t is very large and choose a boundary condition $C(t, S_{max}) = S_{max} - Ke^{-r(T-t)}$, and let $S_t = S_{min}$ if S_t is very low and choose a boundary condition $C(t, S_{min}) = 0$. We can divide $[0, T] \times [S_{min}, S_{max}]$ by grid points and use finite difference method to discretize PDE (13) and then solve a system of linear algebraic equations, see reference books.

5 Call Options Pricing

We summarize the problem we need to solve. Assume risky asset price S follows SDE (11). The payoff at maturity time T is given by $C(T, S) = (S - K)^+$, where K is exercise price. We want to find its price at time 0, that is,

$$C(0, S_0) = E[e^{-rT}(S_T - K)^+], \quad (17)$$

where S_0 is asset price at time 0. The following methods can be used to find $C(0, S_0)$.

1. Closed-form formula (16).
2. Multi-period binomial model with n periods, solve backwards with one period formula and CRR model (7).
3. Finite-difference method to solve PDE (13): use central difference quotients to approximate first and second derivatives.

4. Simulation method to compute expectation (17): generate sample paths of S , evaluate terminal payoff $(S_T - K)^+$, repeat M times to get average, finally multiply discount factor e^{-rT} .

Data. Parameters needed are S_0, r, σ, K, T and n and M . For example: $S_0 = 5, r = 0.05, \sigma = 0.3, K = 5, T = 1$ and $n = 10$ and $M = 10000$.

Tables. List numerical call option prices at time 0, use closed-form formula as benchmark, compute absolute and relative errors; Use different values S_0 while fix other parameters, find average errors and variance.

Figures. Draw call prices at time 0 as a function of S_0 with different methods.

Other possibilities. Compute Greeks with different methods and compare their results.